

Project Initialization and Planning Phase

Date	5 July 2026
Team ID	
Project Title	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to provide an AI-powered platform that helps borrowers manage multiple debts, analyze financial health, and generate settlement strategies — improving financial decision-making, reducing stress, and supporting better debt recovery outcomes. Key features include AI-based financial analysis, settlement prediction, negotiation support, and automated settlement letter generation, with a reliable rule-based fallback ensuring the platform works even without live AI access.

Project Overview	
Objective	The primary objective is to build an AI-powered platform that analyzes financial health, predicts realistic settlement options, and provides intelligent negotiation support to help borrowers achieve debt relief.
Scope	The project includes user authentication, loan management, financial health analysis (DTI/EMI/surplus/stress), AI-based settlement prediction, negotiation letter generation, and a dashboard for tracking loans and settlements.

Problem Statement	
Description	Borrowers struggle with multiple debts, lack of financial guidance, high-interest rates, and limited negotiation support, leading to stress, poor financial health, and risk of default.
Impact	These challenges result in financial instability, damaged credit scores, increased collection calls, and limited options for effective debt relief, affecting both individuals and lending institutions.

Proposed Solution	
Approach	The platform combines a rule-based financial engine with Google Gemini AI to analyze financial data, predict settlement outcomes, suggest negotiation strategies, and generate personalized settlement letters.

Key Features	• AI-powered financial health analysis (DTI ratio, EMI ratio, surplus, stress level) • Settlement prediction and strategy recommendation per loan • AI-based negotiation letter generation (Gemini API + rule-based fallback) • Dashboard for tracking loans, financial health, and settlements • Real-time recalculation for faster, up-to-date debt relief insights • Reliable fallback logic to keep AI features working even without live API access
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Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specifications, number of cores	Standard cloud instance, shared CPU (Render free tier)
Memory	RAM specifications	512 MB (Render free tier backend instance)
Storage	Disk space for data and logs	SQLite file-based database (lightweight, no separate storage tier needed)
Software		
Frameworks	Backend / Frontend frameworks	FastAPI (Backend), React.js with Vite (Frontend)
Libraries	Additional libraries	SQLAlchemy, Pydantic, python-jose, passlib (bcrypt), google-generativeai, pytest
Development Environment	IDE	VS Code, Git, GitHub
Data		
Data	Source, size, format	No external training dataset is required — settlement prediction uses a transparent, rule-based financial engine (DTI/EMI/surplus thresholds) rather than a model trained on public datasets. All data (users, loans, financial profiles, settlements, AI history) is generated live by users and stored in SQLite.